Step 2 : Describe the sample/data. Let $(X_1, X_2, \ldots, X_m)$ be a random sample of m observations drawn from Population-1 and $(Y_1, Y_2, \ldots, Y_n)$ be a random sample of n observations drawn from Population-2, where m and n are small (*i.e.*, $m < 30$ and $n < 30$). Here, these two samples are assumed to be independent.

Step 3 : Set up level of significance (α)

Step 4 : Consider the test statistic

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{m} + \frac{1}{n}}} \text{ under } H_0 \ (i.e., \mu_X = \mu_Y)$$

where S_p is the "pooled" standard deviation (combined standard deviation) given by

$$S_p = \sqrt{\frac{(m-1)s_X^2 + (n-1)s_Y^2}{m+n-2}};$$

and

$$s_X{}^2 = \frac{1}{m-1}\sum_{i=1}^{m}(X_i - \bar{X})^2$$

$$s_Y{}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

The approximate sampling distribution of the test statistic

$$T = \frac{(\bar{X} - \bar{Y})}{S_p\sqrt{\frac{1}{m} + \frac{1}{n}}} \qquad \text{under } H_0$$

is the t-distribution with $m+n-2$ degrees of freedom *i.e.*, $t \sim t_{m+n-2}$.

Step 5 : Calculate the value of T for the given sample $(x_1, x_2, \ldots x_m)$ and $(y_1, y_2, \ldots y_n)$ as

$$t_0 = \frac{(\bar{x} - \bar{y})}{s\sqrt{\frac{1}{m} + \frac{1}{n}}}.$$

Here $\bar{x}$ and $\bar{y}$ are the values of $\bar{X}$ and $\bar{Y}$ for the samples. Also $s_x{}^2 = \frac{1}{m-1}\sum_{i=1}^{m}(x_i - \bar{x})^2$, $s_y{}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$ are the sample variances and $s_p = \sqrt{\frac{(m-1)s_x^2 + (n-1)s_y^2}{m+n-2}}$.

Step 6 : Choose the critical value, t_e, corresponding to α and H_1 from the following table

Alternative Hypothesis (H_1)	$\mu_X \neq \mu_Y$	$\mu_X > \mu_Y$	$\mu_X < \mu_Y$
Critical Value (t_e)	$t_{n-1,\,\frac{\alpha}{2}}$	$t_{n-1,\,\frac{\alpha}{2}}$	$-t_{(n-1),\,\alpha}$

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